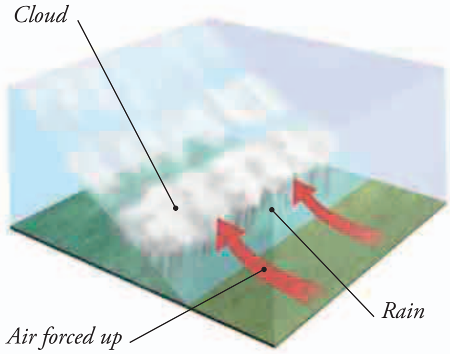
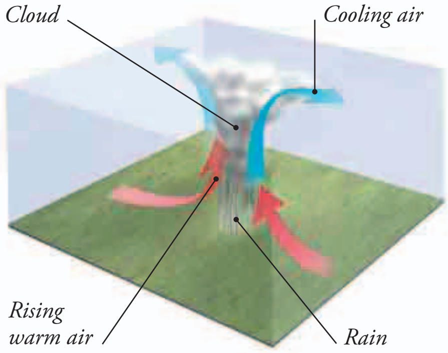
CLOUD FORMATION

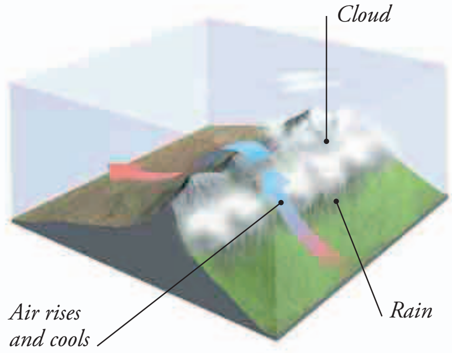
Clouds form when moist air is cooled, often because the air rises and gets colder. There are three main reasons why this can happen, shown below.



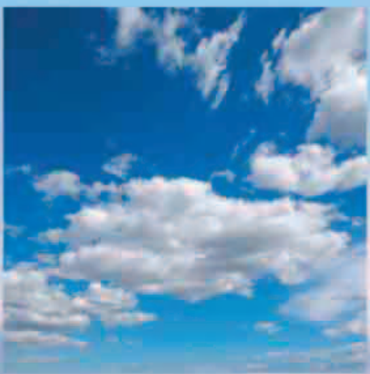
▲ FRONTAL CLOUDS
Clouds form when warm, moist air is pushed up over cold air at a weather front. As the rising air cools, the water vapor condenses into clouds.



▲ CONVECTIVE CLOUDS
When the Sun warms the ground or sea, this warms the air above it. The warm air rises and cools, water vapor condenses, and clouds form.



▲ OROGRAPHIC CLOUDS
If moist air is driven over high ground, it is forced upward. This cools any water vapor in the air, making it condense into clouds.



FAIR WEATHER CLOUDS

When air is warmed, the air molecules move apart so the air gets less dense. This makes the air rise. As it rises and cools, its molecules move closer together, making it denser, so that it stops rising. This point is often marked by a layer of fluffy "fair weather" clouds.

▲ SEA FOG
Fog rolling in from the Pacific Ocean shrouds the Golden Gate Bridge over San Francisco Bay in California.

STORMY WEATHER

When hot sunshine creates a lot of water vapor and warm air, the mixture expands, rises, and cools. The vapor condenses into big convective clouds. This releases heat, warming the air and making it rise higher, carrying more water vapor with it. This then condenses, releasing more heat. This process can create huge cumulonimbus storm clouds that are up to 10 miles (16 km) high.